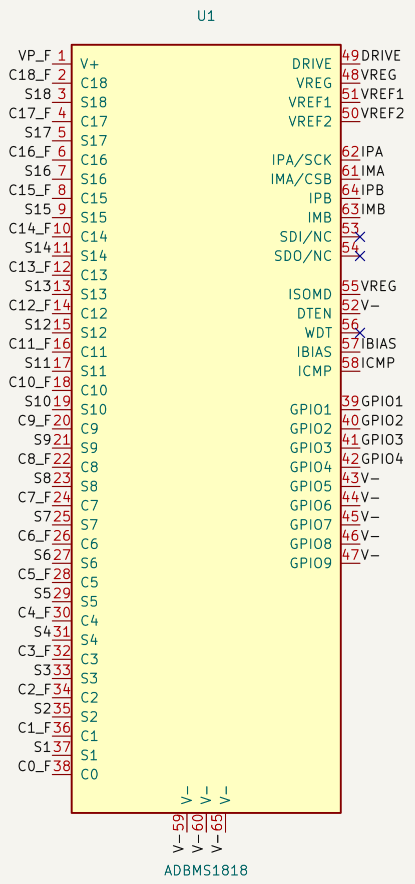
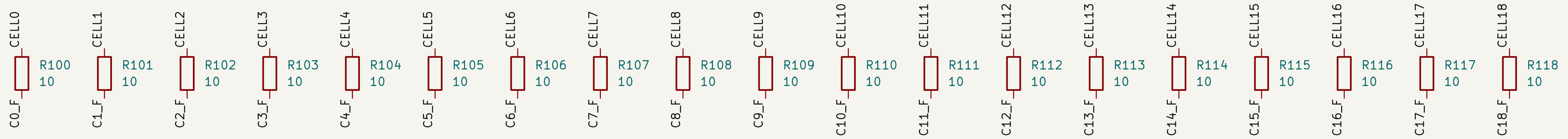
Wired with net labels: labels with the same name are one net. TAPn = connector side of fuse F1nn, CELLn = after it. V− (chip ground) has its own fuse F120.
ADBMS1818 Figure 112, isoSPI filtering per Fig. 104(B). See ../docs/slave−18s−design−basis.md.

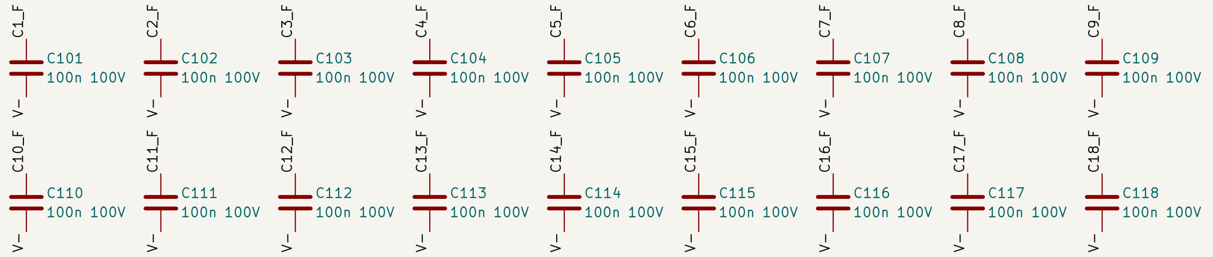
U1 ADBMS1818 (64−LQFP_EP, EP = V−)


U1
ADBMS1818
V+ C18 S18 C17 S17 C16 S16 C15 S15 C14 S14 C13 S13 C12 S12 C11 S11 C10 S10 C9 S9 C8 S8 C7 S7 C6 S6 C5 S5 C4 S4 C3 S3 C2 S2 C1 S1 C0
DRIVE VREG VREF1 VREF2 IPA/SCK IMA/CSB IPB IMB SDI/NC SDO/NC ISOMD DTEN WDT IBIAS ICMP GPIO1 GPIO2 GPIO3 GPIO4 GPIO5 GPIO6 GPIO7 GPIO8 GPIO9
VP_F C18_F S18 C17_F S17 C16_F S16 C15_F S15 C14_F S14 C13_F S13 C12_F S12 C11_F S11 C10_F S10 C9_F S9 C8_F S8 C7_F S7 C6_F S6 C5_F S5 C4_F S4 C3_F S3 C2_F S2 C1_F S1 C0_F
DRIVE VREG VREF1 VREF2 IPA IMA IPB IMB VREG V− IBIAS ICMP GPIO1 GPIO2 GPIO3 GPIO4 V− V− V− V− V−

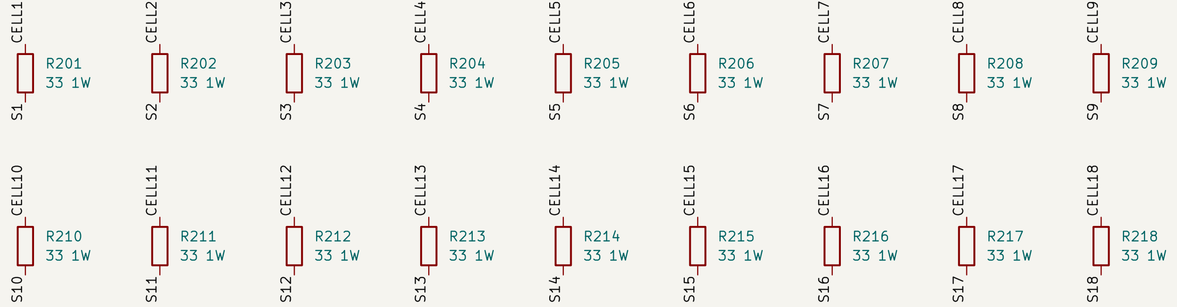
CELL INPUTS R100−R118: 10 ohm series on C0−C18



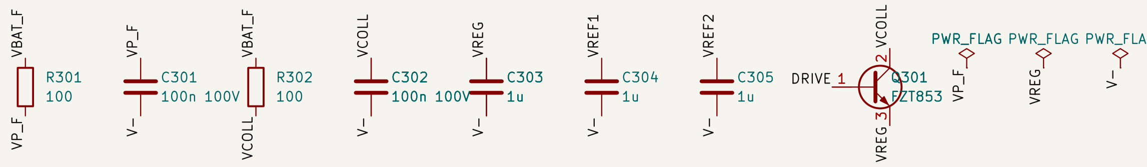
C101−C118: 100 nF 100 V, Cn (after 10 ohm) to V−



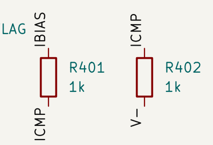
R201−R218: 33 ohm 1 W balance, cell n node (before 10 ohm) to Sn



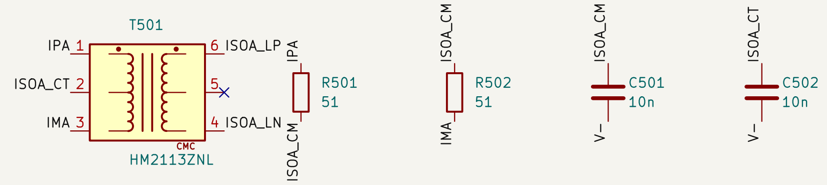
SUPPLY: V+ filter, DRIVE NPN −> VREG, VREF1/2



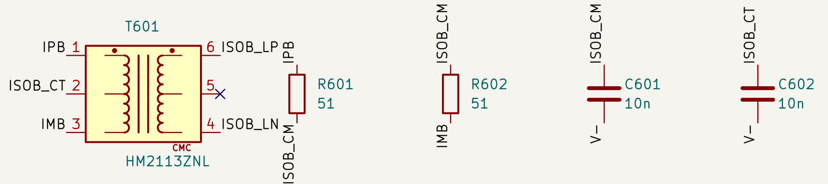
isoSPI BIAS: IBIAS−1k−ICMP−1k−V−



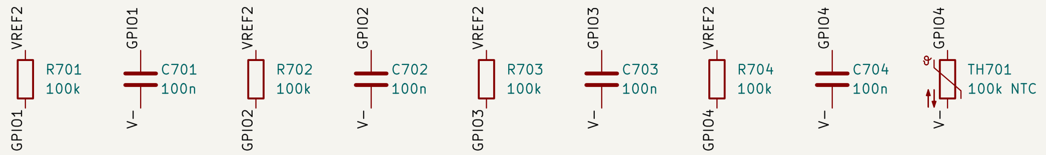
isoSPI PORT A: 2x51 ohm split term + 10 nF CTs



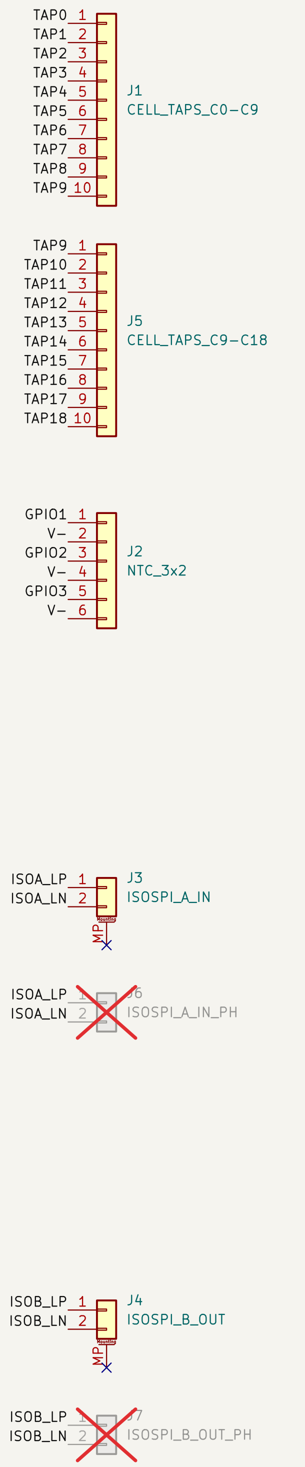
isoSPI PORT B: 2x51 ohm split term + 10 nF CTs



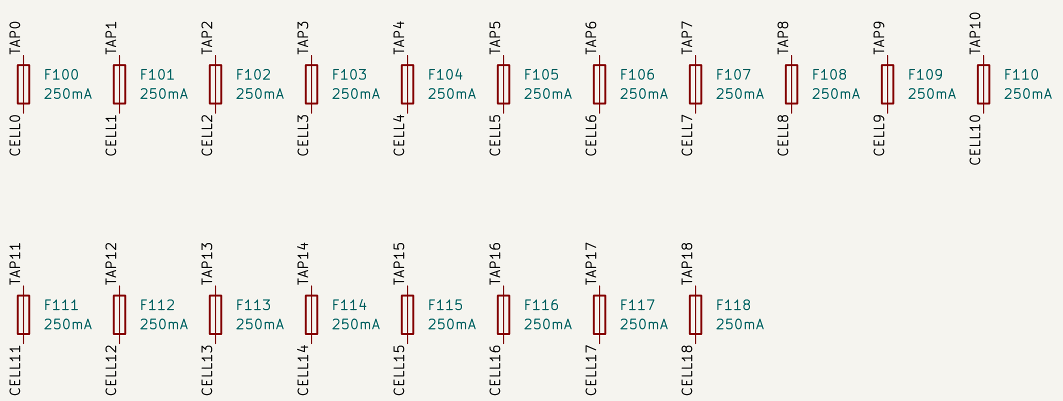
THERMISTORS: 100k to VREF2 + 100 nF, GPIO1−3 cell NTC, GPIO4 onboard



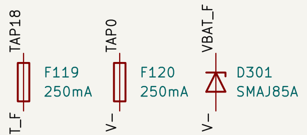
CONNECTORS (JST PH)



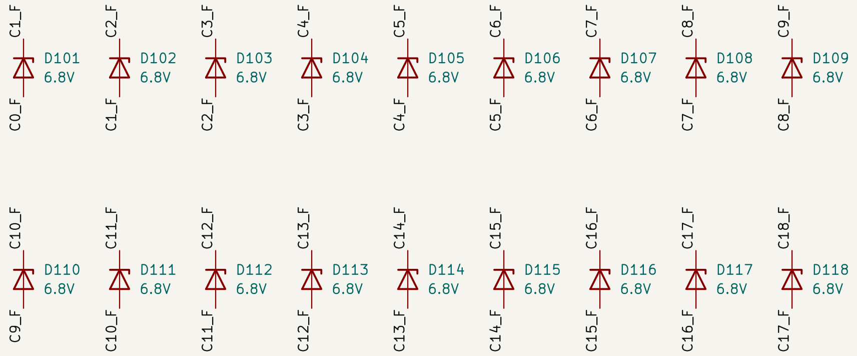
TAP FUSES F100−F118: 250 mA on every cell tap, connector side (AN−2093 Fig. 11)



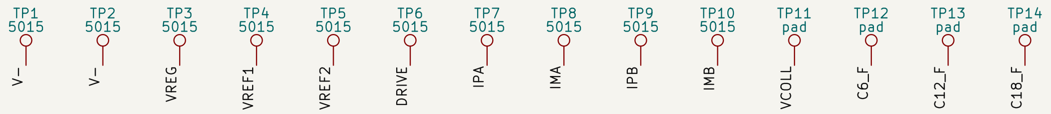
SUPPLY FUSES + TVS F119 top of stack, F120 V−, D301 85 V TVS after F119



ZENER CLAMPS D101−D118: 6.8 V from Cn_F (cathode) to C(n−1)_F, abs max 8 V per pair



TEST POINTS TP1−TP10 Keystone 5015 loops, TP11−TP14 bare 1.5 mm pads (HV, no loop)



MOUNTING HOLES H1−H4: M3, no copper, positions set with the board outline

Not for manufacture
Baseline: ADBMS1818 datasheet Figure 112 typical application

Sheet: /
File: adbms1818−slave−18s.kicad_sch

Title: ADBMS1818 18S slave module

Size: User
Date: 2026−11−06
Rev: 0−draft
KiCad E.D.A. 10.0.6
Id: 1/1